

Research State Briefing

Why did recorded hospitalisations change?

Evidence-supported manuscript and next-study design

Available evidence: count trajectories and exploratory weather models.
Clinical mechanisms, cohort incidence and causal COVID effects remain unresolved.

Sources: receipt, core, journal. Links and scientific limits in notes.

What the present outcome records

Component	Current evidence	Boundary
Population	T2D and/or hypertension diagnosed during 2013-2023	Entry, exits and risk-set construction need query confirmation
Event	First recorded hospitalisation after first CHD/HF diagnosis	Admission cause is unavailable
Scale	132 territory-month counts per outcome	No linked clinical measurements or eligible person-time

An underlying clinical change, a care-use change or an extraction change can alter the same recorded endpoint.

Sources: receipt. Links and scientific limits in notes.

Recent studies raise the novelty threshold

Prior study	Overlapping contribution	Implication
Diabetes, 2025 Hong Kong and Korea	Cardiovascular outcomes, mortality and healthcare use	Another diabetes admissions trend needs a new question
Hypertension, 2025 Hong Kong	Cardiovascular outcomes and blood-pressure control	Adding blood pressure alone does not establish novelty
Care continuity, 2026 Yau et al.	CHD/HF, clinical covariates and competing death	Later follow-up and clinical linkage already have precedent

Candidate contribution: discriminate disease change from observation change.
A novelty gate must confirm that contribution before committing to the expanded study.

Sources: diabetes, hypertension, continuity. Links and scientific limits in notes.

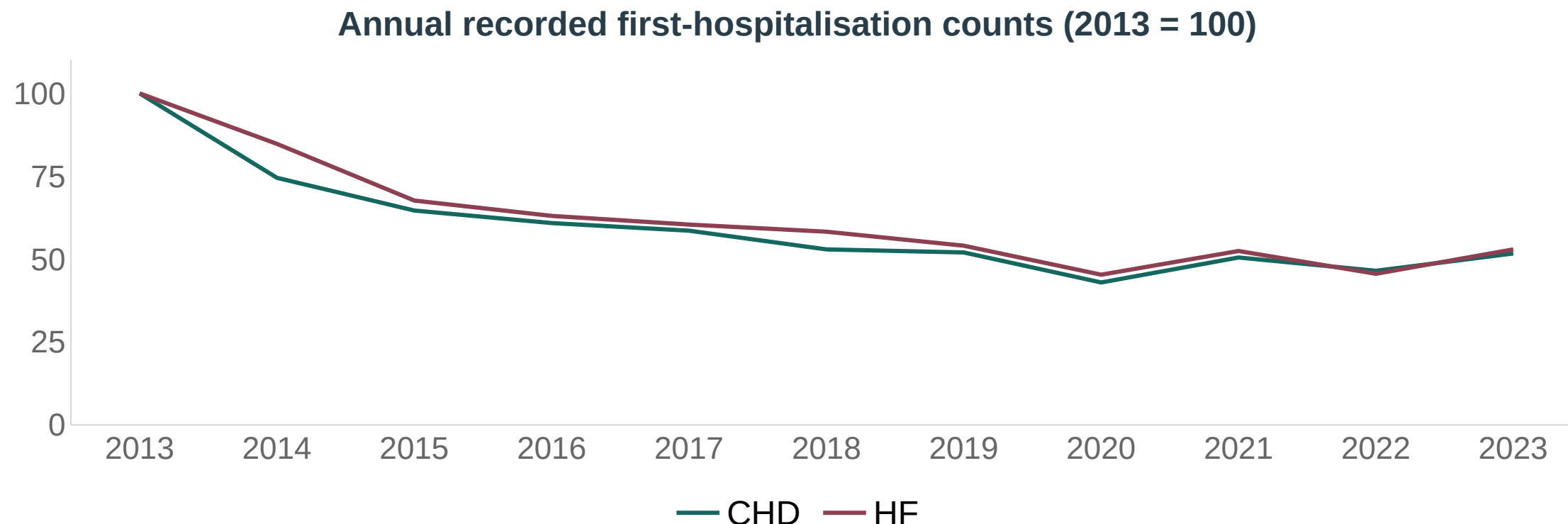
Distinct targets require distinct evidence

Target	Required data	Current status
Recorded burden	Counts with stable extraction definitions	Available, definitions partly unresolved
Cohort incidence	Validated events and eligible person-time	Requires new governed data
Changing weather association	Full-series common basis and direct interaction contrast	Not estimated
Causal health-system mechanism	Defined intervention and defensible identification assumptions	Not identified

Calendar periods describe context. They do not by themselves define a treatment, valid comparator or causal intervention.

Sources: receipt, core. Links and scientific limits in notes.

Recorded counts declined before 2020



Both series were already near half their 2013 totals by 2019.
The 2020 trough and later recovery require a separate explanation.

Sources: annual. Links and scientific limits in notes.

Overlapping windows do not test a period effect

Count ratio per five days	2013-2019 84 months	2013-2023 132 months
HF cold days	1.113 (1.053-1.176)	1.073 (1.006-1.144)
CHD cold days	1.036 (1.007-1.067)	0.995 (0.949-1.043)
CHD hot nights	1.011 (0.991-1.032)	1.022 (1.002-1.042)

These examples use Newey-West lag-6 intervals. The 84 months are nested inside 132, and subsetting changes spline bases and exposure support.

No separate post-period model or formal period-difference test exists.

Sources: sensitivity, core. Links and scientific limits in notes.

Model criticism limits interpretation

Check	Observed issue	Consequence
Calendar control	HF cold-day interval includes 1 under 8 df and year effects	Baseline interval exclusion is fragile
Start-window control	Omitting 24 initial months weakens the HF cold-day estimate	Entry or early extraction remains a rival
Covariance and multiplicity	CHD hot nights depends on SE method. All core BH $q > 0.19$	No protected thermal discovery
Design	Nested window changes mix support, trend and period	A common-basis interaction still needs new execution

The complete released panel remains visible. No preferred fit rescues a headline.

Sources: sensitivity, core. Links and scientific limits in notes.

Rivals make different predictions

Explanation	Discriminating evidence	Failure mode
Improved underlying health	Validated event rates and clinical change beyond selected attendance	Retained attenders alone can create apparent improvement
Reduced or delayed care	Contacts, delayed presentation and mortality linkage	Fewer admissions alone cannot distinguish care from disease
Changed risk set or recording	Entry, prior events, exits and stable event construction	Counts can move without a change in event propensity
Respiratory or thermal context	Matched coverage and formal common-basis contrasts	Adjustment alone does not establish mediation

Multiple explanations may coexist. Severe tests need power to distinguish them.

Sources: receipt, continuity. Links and scientific limits in notes.

The expanded study has a feasibility gate

READY NOW

Exploratory manuscript and source-linked figures
Novelty and measurement audit
Draft governed request and rival predictions

NOT READY FOR A MECHANISM CLAIM

No new transfer or approved model output
No cohort person-time or clinical linkage
No independent validation cohort

Proceed only if verified fields can separate the explanations and the question adds to the closest studies. Otherwise narrow the claim and reconsider journal fit.

Sources: receipt, journal, workflows. Links and scientific limits in notes.

Next sequence and submission boundary

1 DATA AND NOVELTY

Review request, verify schema and close the specific novelty question.

2 DESIGN FREEZE

Prespecify estimands, rivals, error-detection tests and validation before new fits.

3 SECURE EXECUTION

Run in the approved environment. Release reviewed aggregate outputs only.

4 INDEPENDENT CRITICISM

Review identifiability, calibration, missingness and discordant results.

5 MANUSCRIPT AND VENUE

Complete human ethics and authorship facts, then select the journal from evidence.

Sources: journal, workflows. Links and scientific limits in notes.